

Qatar-1b

R-2804

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-1_210815 · created 2026-09-05T00:25:30+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2459441.9408 +/- 0.0021 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.174 +/- 0.056
Transit depth (Rp/Rs)^2	3.04 +/- 1.94 [%]
Orbital Inclination (inc)	80.62 +/- 2.88
Airmass coefficient 1 (a1)	0.86 +/- 0.43
Airmass coefficient 2 (a2)	0.11 +/- 0.4
Scatter in the residuals of the lightcurve fit is	55.13 %
Best Comparison Star	#2 - [518, 51]
Optimal Method	PSF photometry
Transit Duration (day)	0.039 +/- 0.029

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.174 ± 0.056 vs 0.1463 (deviation 0.39σ)
Duration fit vs published	0.039 d vs 0.0692 d (deviation 0.82σ)
Mid-time O–C	3.5 min
Depth SNR	2.4
Residual scatter	55.13 %

Light curve

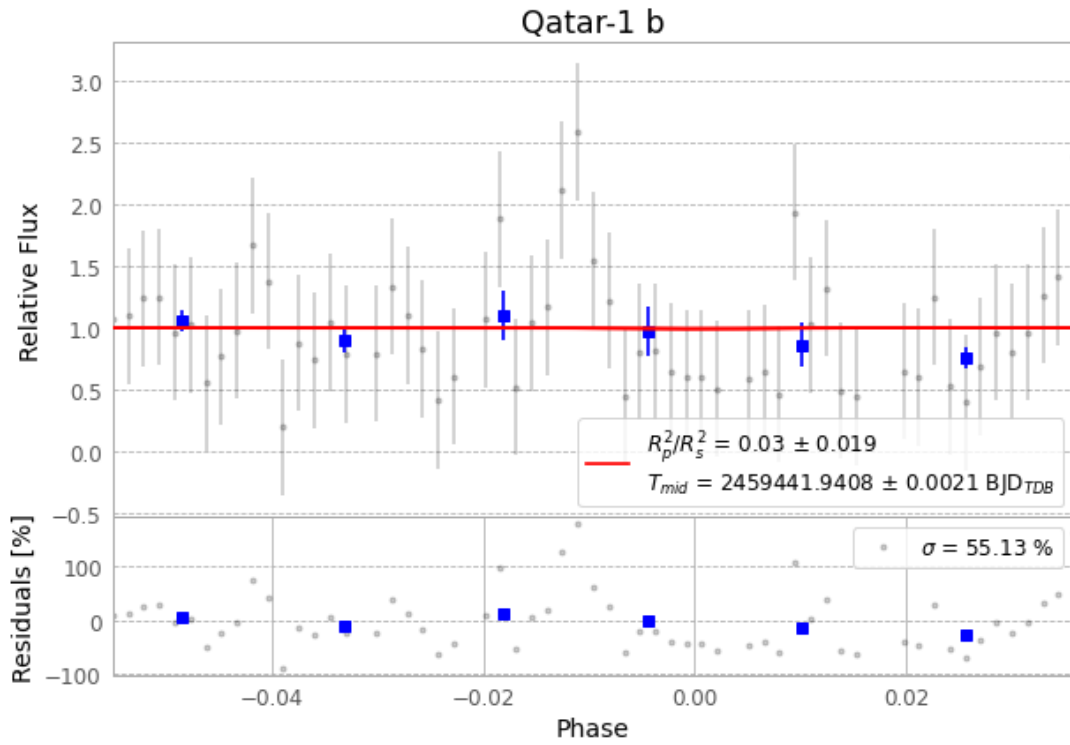


Figure 1 — FinalLightCurve_Qatar-1 b_2021-08-15.png (SHA-256 a319fd5ba065baa1...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [482, 314], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 6dbb4b9e1456570e...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

63 FITS frames (41 MB), Qatar-1210815083912.FITS → Qatar-1210815114512.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-1-210815.log (8025fbd98dd7...), FinalLightCurve_Qatar-1 b_2021-08-15.png (a319fd5ba065...), FinalLightCurve_Qatar-1 b_2021-08-15.csv (2ee1413c77ed...)

Compute resources

Wall time 406.5 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-09-05 00:25Z.